Operating Systems

Mid Term Examination Solutions (E1PY207T)

Section A: OS Functions & Process Management

1. List any four functions of an Operating System.

The primary functions include:

- **Process Management:** Scheduling and managing the execution of processes.
- **Memory Management:** Allocating and deallocating memory space for applications.
- **File Management:** Organizing and tracking files on storage devices.
- **Device Management:** Controlling input/output devices through their respective drivers.

2. Explain the meaning of Deadlock.

A deadlock is a situation where a set of processes are blocked because each process is holding a resource and waiting for another resource acquired by some other process in the set. It results in a permanent halt of execution for those processes.

3. Which scheduling algorithm is best for an interactive time-sharing environment?

Round Robin (RR) is the most suitable. It ensures:

- **Low Response Time:** Each process gets a small unit of CPU time (time quantum) frequently.
- **Fairness:** No process can monopolize the CPU.
- **Interactive performance:** Users receive the first response quickly.

Section B: Schedulers & Resource Allocation

4. Explain the roles of Long-term, Short-term, and Medium-term schedulers.

- **Long-term Scheduler:** Determines which programs are admitted to the system for processing. It controls the degree of multiprogramming.
- **Short-term (CPU) Scheduler:** Selects from among the processes that are ready to execute and allocates the CPU to one of them.

- **Medium-term Scheduler:** Manages swapping processes in and out of main memory to control the degree of multiprogramming or handle memory shortages.

Section C: Deadlock Analysis & Efficiency

5. Deadlock Analysis using Resource Allocation Graph (RAG).

Given: P1 → R2, R2 → P2, P2 → R3, R3 → P3, P3 → R4, R4 → P4, P4 → R1, R1 → P1.
Analysis: This creates a closed cycle: P1 → R2 → P2 → R3 → P3 → R4 → P4 → R1 → P1. In a system with single-instance resources, the presence of a cycle is a sufficient condition for a deadlock. Therefore, the system **is in a deadlock**.

6. Evaluate Process vs. Thread Switching Cost.

Given 25 switches in 1 second:

- **Process Switching:** 25 switches × 4 ms = 100 ms overhead.
- **Thread Switching:** 25 switches × 1 ms = 25 ms overhead.

Conclusion: Thread-based design is significantly better for a web server as it reduces context-switching overhead by 75%, allowing more CPU time for actual request processing.